

# Nikhil Mahajan

Backend Engineer | Java | Spring Boot | AI-Integrated Systems

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🐙 [github.com/ByteBardNikhil](https://github.com/ByteBardNikhil)    🌐 Portfolio    </> LeetCode

## Profile Summary

Backend Engineer with 2+ years of experience in Java, Spring Boot, Microservices, and REST APIs, specializing in enterprise connector versioning and data flow management across cloud and on-premise business platforms on GCP. Skilled in AI-integrated backend systems, LLM automation using Google Gemini, and full lifecycle development from backend services to frontend with React.

## Technical Skills

**Languages:** Java (Java 8 to Java 21), JavaScript, Python

**Backend:** Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, REST APIs, Microservices Architecture

**AI and Automation:** Google Gemini API, LLM Integration, Prompt Engineering, AI Response Evaluation, Python Automation

**Databases and Messaging:** MySQL, PostgreSQL, MongoDB, Redis, Apache Kafka, RabbitMQ

**Cloud and DevOps:** AWS, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, Maven, Git

**Testing:** JUnit 5, Mockito

**Core Concepts:** Data Structures and Algorithms, OOP, Design Patterns, System Design, Distributed Systems, Operating Systems, DBMS, Computer Networks

## Experience

### Backend Engineer – Integration and AI Systems

June 2025 – Present

Bristlecone

Bangalore

- Managed **end-to-end connector version upgrades** by writing JDBC-based validation code, packaging updated JARs, deploying locally via **Docker**, coordinating QA testing, and analysing integration failures and breaking changes before approving version promotion.
- Achieved **90% health check integration success rate** across enterprise connector data flows by validating connectivity, CRUD operations, and data synchronization between GCP and on-premise business platforms.
- Integrated **Google Gemini API** into an internal automation framework for LLM-driven SQL statement analysis – identifying target tools, fields, and operators without cloud execution, reducing analysis overhead by over **80%**.
- Engineered **prompt engineering strategies** to guide Gemini in classifying SQL operations across diverse enterprise connector schemas, improving automation accuracy and reducing manual verification effort.
- Built a **Python-based SQL analysis utility** in Google Colab integrated with Google Drive and Excel, resolving a month-long team bottleneck in **2 days** – recognised publicly by the manager in front of the full team.
- Investigated integration failures and API authentication errors through root cause analysis, identifying breaking changes across connector releases and ensuring uninterrupted data flow for enterprise integrations.

### Software Engineer

April 2025 – June 2025

Tech Mahindra

Ahmedabad

- Maintained enterprise Java applications, resolving **25+ production bugs** during ELTP training program.
- Optimized application performance through code refactoring, improving workflow efficiency by **35%**.
- Collaborated with cross-functional teams to deliver features aligned with business requirements.

### AI Linguistics Trainee

August 2024 – March 2025

Green Rider Technology

Indore

- Evaluated **500+ AI responses** across ChatGPT and Gemini, assessing accuracy, coherence, and contextual relevance to improve model output quality through structured quality ratings.
- Developed evaluation rubrics and annotation guidelines for NLP tasks, contributing to model fine-tuning pipelines and improving response accuracy across live AI projects.
- Built foundational understanding of LLM behavior, prompt sensitivity, and response patterns – directly applied to current AI integration and prompt engineering work at Bristlecone.

### Backend Developer

January 2024 – July 2024

KSV SoftTech

Indore

- Developed a **CRM system using Java and Spring Boot**, managing **1,000+ customer records** efficiently.
- Built a **PDF generation module** automating financial reports, saving **10+ hours weekly** for **50+ clients**.
- Implemented RESTful APIs for CRM integration, enabling seamless data exchange with external systems.

## Projects

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### Orchestrix – Event-Driven Distributed Commerce Platform

*Java 21, Spring Boot, Microservices, PostgreSQL, Redis, Apache Kafka, Docker, React, gRPC, API Gateway*

- Architected a distributed e-commerce platform using Spring Boot microservices with clear service boundaries for User, Product, Inventory, Order, Payment, and Notification domains.
- Designed JWT-based authentication and role-driven authorization for customers, administrators, and sellers.
- Implemented event-driven workflows using Apache Kafka to decouple order processing, inventory management, payment execution, and notification delivery.
- Integrated Redis caching to reduce query latency and improve catalog and product retrieval performance.
- Built resilient inter-service communication using gRPC and REST APIs with fault-tolerance and service isolation.
- Containerized all services using Docker with centralized API Gateway for authentication, routing, rate limiting, and request tracing.

## Education

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### Medicaps University

**Aug. 2020 – June 2024**

*Bachelor of Technology in Computer Science – CGPA: 9.0*

*Indore, India*

## Trainings and Certifications

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### Coding Ninjas

**Feb 2023 – Feb 2024**

*Full Stack Web Development*

- [Full Stack Web Developer](#) | [Backend Development](#) | [Frontend Development](#) | [Core Java](#)

### Udemy

**2024**

*Java and Spring Boot Specialization*

- [Modern Java: Mastering Features from Java 8 to Java 21](#) | [Ninja Slayground 2.0 Level 3 Conqueror](#)

## Achievements

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- Solved **2248+** problems on Coding Ninjas covering DSA, Java, and Web Development.
- Maintained **320-day streak on LeetCode** with 760+ problems solved in Data Structures and Algorithms.
- Winner of **Best Orator** in company-level technical presentation competition at Bristlecone.